CPS607 - Autonomous Mobile Robotics
Assignment 1

General Instructions

Remember to feature your group members' **full names** and **student numbers** prominently on each part of your submission. To avoid suspicions of plagiarism **clearly state any resources (people, print, electronic), the course notes, the textbook, and/or course staff that you consulted** in your submission.

Your submission must consist of:

- A short **written report**, addressing the questions below
- Your **code** implementing the required task on your robot
- A short **video** demonstrating your robot executing the desired task

In addition, each group **must give a brief in-person demonstration to the TA during lab sessions before the deadline** of the assignment. In order to receive full marks, **each member of the group must be present** for the demonstration and be able to participate in the discussion with the TA..

Answer each question completely and justify your reasoning. Your solution will be graded not only on correctness, but also on clarity. Answers that are technically correct but difficult to understand may not receive full marks. Mark values for each question are indicated in **[square brackets]**.

Assignments must be submitted online via the D2L Brightspace Course Shell by the deadline. Late submissions will not be graded. If you require an extension due to accommodations, please contact the lecturer at least three days in advance of the deadline and consult TMU's Academic Accommodation Support. This assignment will be worth 10% of your final grade for this course.

Due Date: October 22, 2026 before 10:00pm

Marking Guide

#1	_____	/ 20
#2	_____	/ 25
#3	_____	/ 55
Total	_____	/ 100

Toronto Metropolitan University

Question 1 - Wheeled Mobile Robot Design

1. Draw a top-down sketch of your mobile robot showing the wheel types and wheel configuration, using the notation and visualization conventions introduced in class. Clearly label all relevant components. For each wheel, specify its type and indicate whether it is steered, powered, or passive.

   [5]

2. Discuss the stability of your robot. Explain whether the robot is stable or unstable, and justify your reasoning by describing the design features that contribute to its stability.

   [5]

3. Discuss the advantages and disadvantages of your robot design in terms of maneuverability and controllability. Identify one key advantage and one key disadvantage overall. Then, analyze the design trade-offs by describing ways to make the robot either more maneuverable or more controllable, and explain the trade-offs associated with each modification.

   [10]

Question 2 - Robot Motion Model and Calibration

1. State the forward and inverse differential kinematics and motion model of your robot, given its physical dimensions (e.g., wheel radius and wheelbase). Assume the robot can be modeled as a differential-drive system. Briefly explain the purpose of each model, what they compute, and what purpose they serve in mobile robotics.

   [10]

2. Your robot is not equipped with wheel encoders or other sensors to measure wheel rotational speed. Instead, the robot is controlled using pulse-width modulation (PWM) duty cycles. To correctly apply the motion model, it is therefore necessary to estimate the relationship between PWM commands and wheel angular velocity. Propose and implement a method to determine an approximate mapping between PWM duty cycle and resulting wheel velocity, assuming a linear relationship. Then, use this mapping to modify your forward kinematics and motion model so that they accept PWM commands as inputs instead of wheel velocities.

   [15]

Question 3 - Open-Loop Path Tracking

1. Figure 1 shows a reference path for your mobile robot. Based on the differential kinematics model from Question 2 and the concepts covered in class, implement an open-loop (i.e., sensor-free) motion sequence that enables your robot to follow the given path as accurately as possible. Begin with the physical model previously derived, but you may refine or recalibrate the model to improve execution accuracy. After completing the path once, the robot should automatically begin another iteration (i.e., run in a continuous loop).

   Briefly describe your overall approach. What steps are required to convert the reference path into executable code and corresponding PWM motor commands? If applicable, explain any modifications or calibrations made to the kinematic model.

   As part of your submission, include a video demonstrating your robot executing the trajectory. The video must show at least three attempted laps of the robot following the route. Additionally, submit the full source code used to generate and execute the motion.

Note: It is expected that the robot will not follow the desired path perfectly (or even particularly well). Tracking accuracy will not be graded strictly. Instead, grades will emphasize correct understanding and application of the methodology, effort, and the quality of the description of your approach, rather than absolute motion precision. Deviations and limitations will be discussed in the next question.

[30]

2. Qualitatively assess how well the robot tracked the desired path. Describe the challenges encountered during implementation, how you addressed them, and the likely sources of error or limitations that affected accuracy. Suggest how these limitations could be mitigated in future if both hardware and software could be modified.

[10]

3. Demonstrate your implemented approach to the TA during lab hours, again showing at least three laps of the path. The grade will be composed as follows: 10 points for successfully running the robot and executing the task (within the expected limitations), and 5 points for demonstrating understanding through a brief explanation of the system to the TA.

[15]

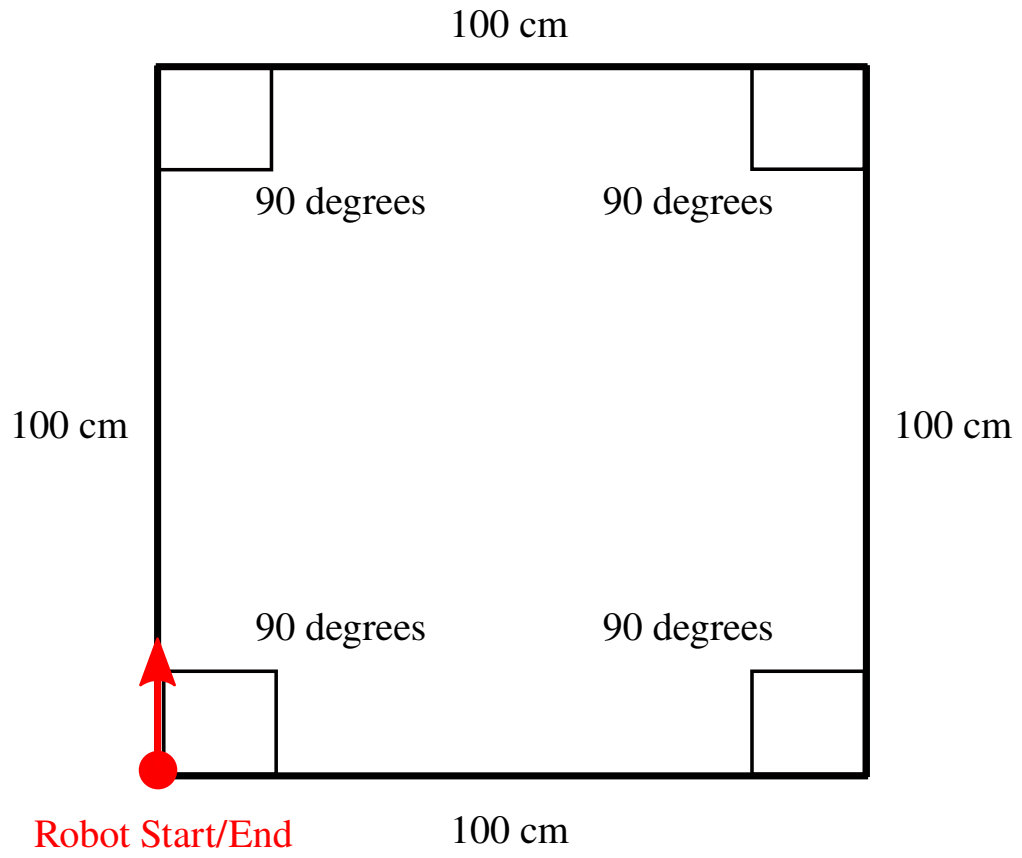


Figure 1: The robot reference path is completely symmetric. Each straight segment is 100 cm long, and the corners feature angles of 90 degrees. The robot starts in the bottom left corner.